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Research Interests

Computational cancer biology; statistical and AI-based methods for single-cell and spatial omics; cellular pathway activity inference; tumor microenvironment modeling; cross-modal learning; colorectal cancer heterogeneity and precision oncology.

Appointments

2023– Postdoctoral Researcher, Department of Biostatistics and Bioinformatics, Duke University. Mentor: Jichun Xie.

Education

2017–2023 Ph.D. in Computer Science, Tongji University, Shanghai, China. Advisor: Kai Yang.
Anomaly Detection, Early Prediction, and Diagnosis of Large-Scale Computing Systems Using Log Data Analysis

Research Funding

2026–2031 NIH/NCI K99/R00 Pathway to Independence Award (K99CA312801). Role: PI.
Pathway-Centric AI for Dissecting Colorectal Cancer Heterogeneity Using Single-Cell and Spatial Omics. Total: \$971,395.
Mentors: Jichun Xie (primary), Andrew Nixon, John Strickler, Xiao-Fan Wang.

2024 Duke Center for Human Systems Immunology Pilot Award. Role: PI. \$20,000.
TEMPO: Prediction of Tumor Immune Microenvironment Subtypes Using H&E Image Data Alone with AI-Enhanced Cross-Modal Transfer Learning from Genomic Data.

Ongoing Research

Programs	Developing statistical and AI-based methods for cellular pathway activity inference and spatial heterogeneity in colorectal cancer (SPACT and SPACT-ST).
Virtual cells	Investigating context-associated changes in cell states and tissue microenvironments through conditional virtual cell modeling.
Translation	Exploring cross-modal models to relate molecular and spatial programs to pathology images and clinically accessible cancer measurements.

Publications

* Denotes equal contribution.

Full list available on [Google Scholar](#).

Journal and conference papers

- [1] **Xie, Y.**, & Yang, K. *Log Anomaly Detection by Adversarial Autoencoders with Graph Feature Fusion*. IEEE Transactions on Reliability, 73(1):637–649 (2024).
- [2] Jiang, X., **Xie, Y.**, & Xie, J. *xMINT: A Multimodal Integration Transformer for Xenium Gene Imputation*. ICML Workshop on Accessible and Efficient Foundation Models for Biological Discovery (2024).
- [3] **Xie, Y.**, & Yang, K. *Domain Adaptive Log Anomaly Prediction for Hadoop System*. IEEE Internet of Things Journal, 9(20):20778–20787 (2022).
- [4] **Xie, Y.**, Yang, K., & Luo, P. *LogM: Log Analysis for Multiple Components of Hadoop Platform*. IEEE Access, 9:73522–73532 (2021).
- [5] Zhang, C., Xie, T., Yang, K., Ma, H., **Xie, Y.**, et al. *Positioning Optimization Based on Particle Quality Prediction in Wireless Sensor Networks*. IET Networks, 8(2):107–113 (2019).
- [6] Feng, M., Huang, W., Wang, Y., & **Xie, Y.** *Multi-Organ Segmentation Using Simplified Dense V-Net with Post-Processing*. SegTHOR Challenge, IEEE International Symposium on Biomedical Imaging (ISBI) (2019).

Manuscripts under review and in preparation

- [7] **Xie, Y.***, Ferrer, O.*, & Xie, J. *Using Gene Co-expression Graph Topological Fingerprints to Enhance Pathway Activity Scoring*. Under review at Genome Biology (2026).
- [8] Yan, Z.*, **Xie, Y.***, Ferrer, O., Xie, J., & Velmeshev, D. *Meta-tracker: A Machine Learning Framework to Project and Align Lineage Dynamics by Leveraging a Comprehensive Atlas of Reference Trajectories*. In preparation.

Patents

- [9] Yang, K., & **Xie, Y.** *Method for Diagnosing the Causes of Hadoop System Anomalies Using Graph Analysis*. China Patent ZL201910318337.X (granted).
- [10] Yang, K., & **Xie, Y.** *Method for Log Pattern Extraction and Matching*. CN110879802A.
- [11] Yang, K., & **Xie, Y.** *Anomaly Detection Method Based on Deep Siamese Autoencoder Networks*. CN114726581A.
- [12] Yang, K., **Xie, Y.**, et al. *Log Anomaly Detection Method Based on Adversarial Transfer Learning*. CN116010208A.
- [13] Guo, Y., **Xie, Y.**, et al. *Data Anomaly Detection Method, System, and Electronic Device*. CN115203186A.

Software

xGATE	Topology-aware pathway activity scoring for single-cell and spatial data. github.com/jichunxie/xGATE
xMINT	Multimodal H&E and gene-panel imputation for Xenium data. github.com/jichunxie/xMINT_manu_support

Invited Talks and Presentations

Aug 2026	<i>Meta-tracker: A Machine Learning Framework to Project and Align Lineage Dynamics by Leveraging a Comprehensive Atlas of Reference Trajectories</i> . Poster, Gordon Research Conference on Neural Development, August 2–7, 2026.
Sep 2024	<i>A Robust and Accurate Test for Cellular Pathway Activities</i> . Invited speaker, 5th Annual Emerging Scholars in Genome Sciences Symposium, University of Virginia.
Jul 2024	<i>xMINT</i> . Poster, ICML Workshop on Accessible and Efficient Foundation Models for Biological Discovery, Vienna, Austria.
May 2024	<i>Leveraging AI to Link Pathology Images with Genomic Data for Enhanced Clinical Insights</i> . Invited talk, Duke CHSI Retreat Symposium.
Aug 2023	<i>AIOps and Its Future</i> . Invited talk, The Chinese University of Hong Kong, Shenzhen.

Teaching

Fall 2022	Graduate Teaching Assistant, <i>Data Analysis</i> , Tongji University.
Spring 2022	Teaching Assistant, <i>Data Collection and Integration Technology</i> , Tongji University.
Fall 2021	Teaching Assistant, <i>Web Technology Basics</i> , Tongji University.
Fall 2018	Teaching Assistant, <i>Information Security Theory and Technology</i> , Tongji University.

Student Mentoring

2026	Tianning Yang, M.S. student, Biostatistics and Bioinformatics, Duke University. Project: T-cell-based biomarkers linked to clinical outcomes.
2025	Tesko Chaganti, undergraduate, Johns Hopkins University (SenNet Consortium Undergraduate Student Program intern, co-mentored with the Nixon lab). Project: Tissue-based heterogeneity of cellular senescence.
2025	Xinyue Yang, M.S., Biostatistics and Bioinformatics, Duke University. Project: Single-cell RNA-seq analysis.
2023–2025	Yixuan Shao, M.S., Biostatistics and Bioinformatics, Duke University. Project: Analysis of glioblastoma Xenium data. Now: Biostatistician, LLX Solutions.
2023–2024	Yaoyao Li, M.S., Biostatistics and Bioinformatics, Duke University. Project: Computational pathology imaging analysis with HTAN datasets. Now: Ph.D. student, Rutgers University.

Honors

2026	NIH/NCI K99/R00 Pathway to Independence Award.
2025	NIH Compass Program Fellow.
2024	Invited Speaker, Emerging Scholars in Genome Sciences Symposium, University of Virginia.
2023	Outstanding Graduate of Shanghai.
2014, 2015	National Scholarship, Ministry of Education of China.
2013–2017	First Prize Scholarship (seven times), Qingdao University.

Professional Development

2026	NCI Career Development Workshop: Advancement to Research Independence.
2026	Design and Management of Cancer Clinical Trials Course.
2025	NIH Compass Training and Mentoring Program, Washington University School of Medicine.
2025	Boot Camp for AI and AI-Accelerated Medical Research, Duke University.
2024	AI+Science Summer School, University of Chicago.

Reviewing

IEEE Internet of Things Journal; IEEE Journal of Biomedical and Health Informatics; IEEE Transactions on Consumer Electronics; Frontiers in Oncology; Frontiers in Immunology.